

Research on Intelligent Modeling Analysis and Recognition System in Architectural CAD Engineering Drawing

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Abstract—In mechanical design, this paper uses CAD interface for 3D modeling to generate 3D solids. This paper clarifies the intelligent standard view generation technology in 3D CAD software and its intelligent technologies such as axonometric drawing, internal structure, auxiliary view, intelligent annotation, parametric design and assembly design, showing its powerful application function. According to the principle of shape design in 3D conception, this paper discusses the teaching mode of combining composition theory with 3D CAD application, and using 3D CAD system to cultivate students' abstract thinking, image thinking and innovative thinking.

Keywords—architectural CAD, engineering drawing, intelligent modeling, recognition system

I. INTRODUCTION

In the context of the Internet era, the practical application of CAD technology and computer technology can improve people's quality of life. Using CAD technology for architectural drawing design can express the key content of construction projects, make the practice and development of architectural drawing more theoretical, scientific and systematic, and help modern urban planning and design [1]. The application of CAD technology has gradually replaced the traditional architectural drawing method, playing an active role in the actual building construction operation, and improving the efficiency of architectural drawing.

II. OVERVIEW OF CAD DRAWING TECHNOLOGY

Auto CAD is the abbreviation of Auto Computer Aided Design, which is a widely used technical form. CAD drawing technology is relatively simple to operate, and has a wide range of adaptability and interactivity. In the specific application stage, it needs to be combined with control charts, binary graphics and basic graphics, and interactive applications are carried out through commands or menu selection. CAD drawing technology can provide a variety of graphic design drawings for the project. In the design stage, it needs to be based on diversified and intelligent systems, and the drawing technology should be reasonably applied in different fields. Mechanical drawing is usually a drawing provided to workers before the machine is produced, including the size, size and structural style of the machine [2]. The design form needs to be analyzed and carried out according to the construction requirements. Mechanical drawing is also an important bridge for communication between designers and manufacturers. It is necessary to clearly express various parameters and requirements to avoid psychological obstacles to manufacturers and affect the progress of engineering projects. The drafting design

itself includes axonometric drawings, schematic diagrams and parts drawings, etc., which should be carried out according to the requirements.

At present, 3D drawing software based on AR technology has been extended to the teaching practice of engineering drawing and CAD courses. This kind of software has good virtual reality and can "seamlessly" integrate the real world and the data world, expand the available range of teaching scenarios, and improve the classroom teaching effect of engineering drawing and CAD courses. This kind of software can be used not only on computer terminals, but also on mobile terminals. The operation is not complicated, and it is simple and easy to use, which enriches the content of engineering drawing and CAD courses [3]. The design and development of AR technology applications in smart terminal equipment provides convenient conditions for students to practice and map anytime, anywhere, greatly increases the extensibility of school classroom teaching, and creates a good environment for students to learn. The application of AR technology in engineering drawing and CAD course teaching has improved the application level of virtual three-dimensional models, and cooperated with JPCT-AE and Unity3D three-dimensional engines to generate three-dimensional three-dimensional shapes from flat graphics, replacing wooden molds and wall charts, and promoted basic teaching equipment. The quality of mechanical drawing has been improved, and the teaching effect of mechanical drawing has been improved.

III. 3D SOLID MODELING METHOD

Engineering drawings are the primary tools for engineers and technicians to describe design objects and express design ideas. They are essential technical support for product design and production. They are widely used in the field of mechanical parts design. Since Idesawa first published a paper on the reconstruction of 3D solids from engineering three views in the early 1970s, research on this topic has received widespread attention. So far, many articles in this area have been published, and this paper can roughly divide them into (1) Geometric element layered modeling method, (2) Model-guided recognition modeling method, (3) Body cutting-based modeling method.

A. Hierarchical modeling method of geometric elements

The hierarchical modeling method of geometric elements is to gradually establish three-dimensional points, lines, surfaces, and volumes according to drawing rules and the corresponding relationship between two-dimensional points and lines. The main steps are as follows: calculate

three-dimensional vertices from two-dimensional points; find three-dimensional edges from three-dimensional vertices; The faces that make up a three-dimensional body; the three-dimensional faces make up a three-dimensional solid.

(1) Idesawa proposed a mathematical processing form for three orthographic views of a three-dimensional entity. In constructing 3D vertices, 3D line segments, and 3D surfaces, it is found that there are points that cannot exist. The algorithm applies some discriminative rules to eliminate these components in the magic map. For example, as a 3D vertex, it should belong to at least three 3D edges; and for a 3D edge, it should belong to at least two non-coplanar faces. The algorithm is only suitable for polyhedrons and cannot handle ill-conditioned knots and multi-solution problems.

(2) After enriching the results of the previous research, the authors of Preiss proposed a method that can generate plane and cylindrical surfaces.

Algorithms for three-dimensional shapes. The algorithm directly uses the ring on the view to generate the visible 3D surface. The main idea is to impose constraints on aggregation, the so-called constraint propagation. It does not generate 3D graphics primitives that do not meet the conditions to limit the algorithm's search space. It consists of four steps: by imposing constraints on the vertices, weave out the candidate vertex list; by imposing constraints on the edges, weave out the candidate edge list; by testing every two edges, delete the incompatible edges; Constraints are imposed on edges in and between rings and rings, and edges form rings; the algorithm still cannot handle multi-solution and ill-conditioned solution problems.

B. Model-guided recognition modeling method

From the perspective of constructing solid geometry, any complex shape can be decomposed into a combination of some predefined three-dimensional shapes. Suppose the three-dimensional features of voxels and their three-view projection feature descriptions can be defined as a series of patterns to form a voxel recognition model. In that case, the input view graphics can be classified according to various model requirements and matched using different patterns; then, when The graphics in the view match the three-view mode of a voxel model, the corresponding sub-volume can be directly constructed using its three-dimensional view mode. This is the model-guided reconstruction method. This method avoids the considerable search and elimination operations of three-dimensional points and lines in the bottom-up method, improves the speed, and has a simpler algorithm.

C. Modeling method based on volume cutting

The above reconstruction method starts from the two-dimensional graphic data and gradually deduces the structure of the three-dimensional shape. The reconstruction method based on volume cutting starts from the three-dimensional shape, and through the continuous cutting of the obtained entity, the three-view projection gradually approaches or even completely matches the three-view to be identified. The characteristic of the reconstruction method based on volume cutting is that it always operates on the volume, surface, and edge of the three-dimensional space and only judges and cuts the generated possible false sub-volumes [4]. However, this method brings a large number of complex three-dimensional space manipulation calculations.

IV. CAD/GIS INTEGRATION SCHEME DESIGN AND IMPLEMENTATION

A. CAD/GIS Integrated Framework Design

Research the spatial data access technology of CAD software (AutoCAD) and GIS engine (GDAL), realize the internal integration of CAD/GIS, and process data in CAD and GIS format uniformly. The system framework design is shown in Figure 1 (the picture is quoted from Geographic Information Systems Architecture).

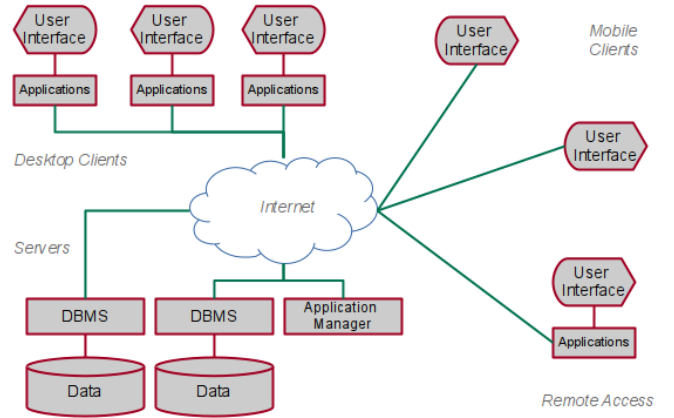


Fig. 1. Schematic diagram of software architecture design

The system is mainly divided into three layers. Data layer: Add support for GIS data on the basis of AutoCAD data, can associate CAD entities and GIS elements, edit and save graphics and attributes synchronously; business layer: embed GIS engine GDAL through Object ARX, and on this basis Increase the basic functions of GIS, introduce the GIS data management mechanism, and realize the data integration management of AutoCAD and GIS; Engineering geological section, house distribution map, tower base shape and section, geographic coordinate system, geographic image/satellite map and other functional modules.

B. CAD/GIS Dynamic Symbology

Dynamic symbols are the key to data management and graphics drawing. The software uses ObjectARX custom entities and OGR vector elements in GDAL to define symbols and associate related parameters (as shown in Figure 2, the picture is quoted in An Extended Symbol Table Infrastructure to Manage the Composition of Output-Specific Generator Information). This method cleverly integrates "GIS Feature Data Model" with "CAD Graphics Drawing" while maintaining the integrity of symbolic objects. Symbol objects are displayed dynamically according to parameters such as scale, without external intervention.

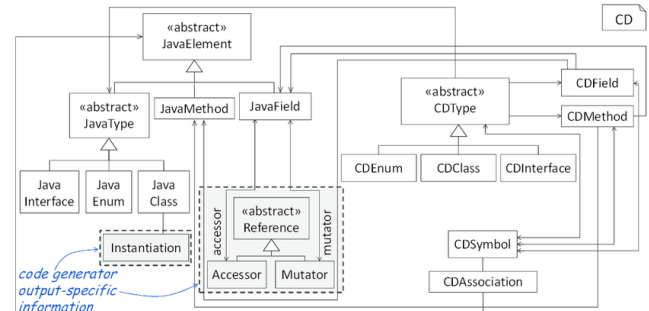


Fig. 2. Schematic diagram of symbol system structure design

The host platform AutoCAD manages the GIS/CAD integrated entities through the original database management mechanism, and at the same time calls the GIS engine to manage the "GIS data" of the integrated entities, thereby constructing a "synchronized" and "heterogeneous" management mechanism for the symbol system (as shown in the figure 3), the picture is quoted from Wireless Monitoring System of Household Electrical Consumption with DALY-based Control Unit of Lighting Facilities Remotely Controlled by Internet.

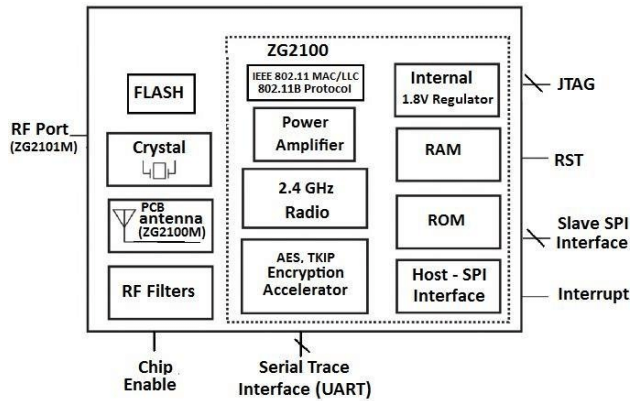


Fig. 3. Host platform AutoCAD

C. Structure and function design of engineering drawing collaborative learning support module

Synergy between members of a set of synergies, usually done through interaction. According to the spatial location and response mode of the interaction mode, the working modes of collaborative learning can be divided into four categories: face-to-face interaction, asynchronous interaction, asynchronous distributed interaction and synchronous distributed interaction [5]. Among them, face-to-face interaction is the simplest, usually in the form of a meeting; asynchronous interaction can be realized through a shared database; asynchronous distributed interaction requires the support of the network, which can be realized through file management, E-mail, distributed database, etc.; synchronous distributed interaction It is difficult to implement. In addition to various support tools required for asynchronous distributed interaction, an environment that can support real-time collaboration is also required. In order to provide a natural collaboration platform for those who participate in collaborative learning, the collaborative learning system must support these interactive ways.

CSCL support system mainly refers to some collaborative support modules on the server side in the CSCL system structure, including collaborative session support module, collaborative map recognition support module, collaborative drawing support module and intelligent control support module, to support the entire collaborative learning environment.

a) Collaborative conversation support module: Collaborative conversation mainly refers to the sharing of text information, providing a collaborative environment for each collaborative learning member to express opinions and discuss on a certain topic. The function of this module can be realized by, for example, "whiteboard system". The key here is to implement functions such as collaborative session

control, access control, and floor control.

b) Collaborative graph recognition support module: Collaborative graph recognition mainly refers to the way of realizing WYSIWIS (what you see is what I see) between collaborative members in a collaborative learning environment. The key here is how to transmit and communicate the vector graphics which are widely used in this system among the collaborative learning members through the network and display them on each collaborative learning member side [6]. Because the vector data itself has carried out high-strength compression on the graphic data, the data redundancy is extremely small, which can reduce the communication volume and improve the network speed.

c) Collaborative drawing support module: Collaborative drawing mainly accepts and forwards drawing messages of collaborative learning members, and ensures that the data information in the views of collaborative learning members is consistent. The key here is to realize that the Web server can control the special application software of the system learning members accordingly, accept the actions of the collaborative learning members and forward them to other collaborative members in the collaborative learning environment.

d) Intelligent control support module: The intelligent control module mainly adds some intelligent control functions to the CSCL system, such as evaluating the learning effect of the collaborative learning members and intelligently judging the answering questions. The key here is to establish a suitable evaluation model, including the establishment of a reasonable data structure for students and question banks, so as to have certain intelligent control over the entire collaborative learning process.

V. SYSTEM PRODUCTION AND INTELLIGENT IDENTIFICATION

A. Intelligently generate auxiliary views

In the field of engineering drawing, in addition to using a large number of standard views in the six directions specified by national standards, in order to clearly express the internal and partial structure of the part, it is necessary to use the form of direction view, partial view, partial enlarged view and section view. Auxiliary view. The 3D CAD software also provides an intelligent method for the generation of these views [7]. Taking the section view as an example, the generation methods of various types of section views such as full section, half section, stepped section, and unfolded section are basically similar. Specify the position of the section line in the basic view and select the corresponding command button, the software can automatically generate the required section view.

B. Observe the internal structure of cavity parts

For parts with complex internal structures, 3D CAD software can cut the entire part so that the internal structure can be clearly displayed, so that part designers and manufacturing workers can clearly see the internal structure of the part. Figure 4 shows the display effect of cutting the tee parts in the CAXA solid design. From the figure, the stepped hole structure inside the cylinder can be clearly seen.

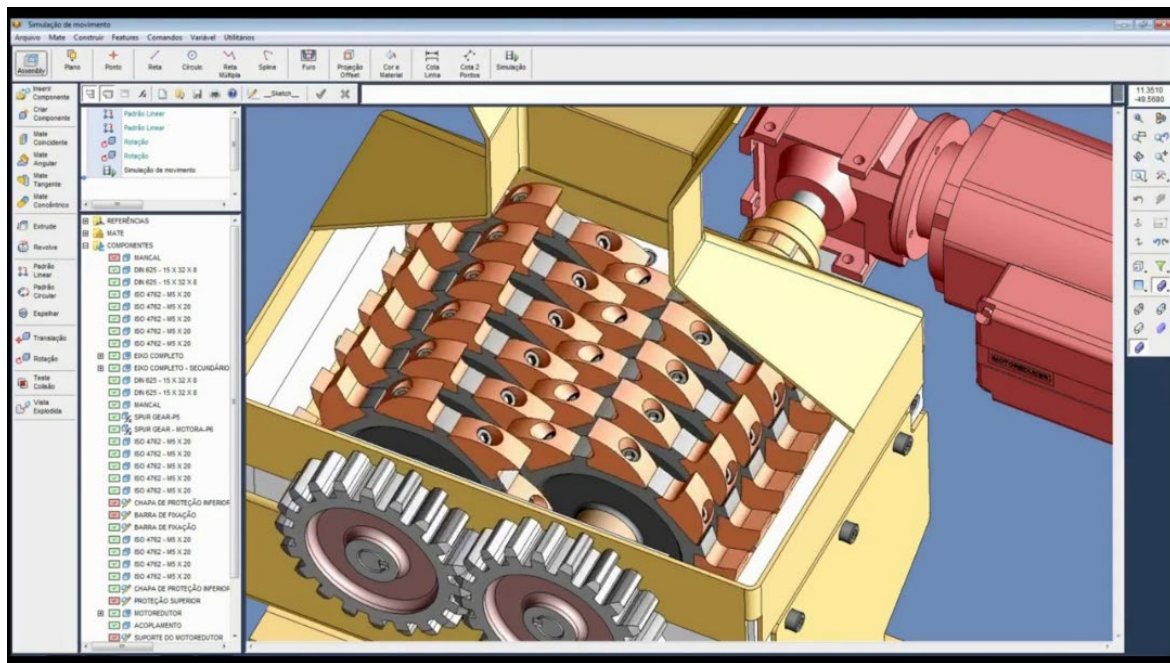


Fig. 4. Intelligently generated tee cut effect

VI. CONCLUSION

The basic form design is different from the product design. It is a kind of transitional training that does not need to consider the use environment, practical requirements, production process, and manufacturing cost of the entire product. All that needs to be done is to study the essential elements that form the three-dimensional form of products so that students can not only study and design in depth around a particular aspect of a specific product in the three-dimensional form design but also have a broad perspective design. Moreover, accessible space to exert imagination and creativity. Starting from the CAD 3D object composition analysis, the existing CAD modeling methods are summarized and abstracted into two easy-to-understand categories: basic modeling and complex modeling. A method of comprehensively considering the essential elements of morphological composition and CAD 3D modeling is proposed, and the concept of design is introduced into teaching through examples.

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